Q.26 Write a short note on fuzzy logic

Q.27 Explain limit cycle and dead zone.

Q.28 Write a short note on single loop and multi loop control system?

Q.29 What is degree of freedom in robotics? Explain the robot arm configuration.

Q.30 Write down the advantages and disadvantages of cascade control.

Q.31 Write a short note on robotics.

Q.32 Write a Labview software.

Q.33 Discuss about the hysteresis and backlash.

Q.34 Discuss about the artificial intelligence.

Q.35 Explain briefly about the MATLAB software.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Discuss feed forward and cascade control. Write the differences between feedback and feed forward control system

Q.37 Write a short note on :

a) Artificial intelligence

b) Fuzzy logic

c) Inherent non linearities

d) Friction

Q.38 What are the different non linearities? Explain linear and nonlinear system. Write any four differences between linear and nonlinear system

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Roll No. ....

4th Sem / IC

Sub : Advanced Control System

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 _____ occurs when a transducer with low damping is driven near its resonance frequency, resulting in sudden increases or decreases in oscillations

a) Jump resonance b) Jump phenomena

c) Limit cycle d) None of the above

Q.2 Which of the following control system detects the disturbance after the controller?

a) Feed forward b) Feedback

c) Cascade d) Split range

Q.3 ______ is the deliberate introduction of non-proportional, nonlinear components into a system to enhance performance, increase safety, or improve efficiency compared to purely linear designs.

a) Intentional b) Backlash

c) Inherent d) Hysteresis

Q.4 _____ is a clearance or lost motion in a mechanism caused by gaps between the parts

a) Hysteresis b) Limit cycle

c) Backlash d) Dead zone

Q.5 _____ a branch of computer science that develops systems capable of simulating human cognitive functions, such as learning, reasoning, problem-solving, perception and decision-making.

a) Fuzzy logic
b) Backlash
c) Artificial intelligence
d) Neural network

Q.6 _____ is a subset of machine learning that mimics the human brain's structure to process data, identify complex patterns and make predictions

a) Degree of freedom b) Neural network
c) Fuzzy logic d) All of the above

Q.7 DOF stands for

a) Degree of friction b) Degree of freedom
c) Direction of freedom d) Direction of friction

Q.8 Which of the following is not a nonlinear characteristic of the system?

a) Hysteresis b) Precision
c) Jump phenomena d) Backlash

Q.9 Which of the following is the special type of cascade control.

a) Ratio control b) Feed forward control
c) Split range control d) Ratio control

Q.10 _____ is the dependence of a system's current state on its history, where the output lags behind the input.

a) Backlash b) Hysteresis
c) Limit cycle d) Friction

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Expand DOF.

Q.12 Define multi loop control.

Q.13 A ____ control is a proactive, model-based strategy they measures disturbances directly to adjust manipulated variables before they affect the output.

Q.14 Write any one application of split range control.

Q.15 Backlashis a clearance or lost motion in a mechanism caused by gaps between the parts. (T/F)

Q.16 Give any one example of nonlinear system.

Q.17 What is degree of freedom?

Q.18 _____ is a form of many valued logic that mimics human reasoning by allowing for degree of truth than strict binary.

Q.19 What is SCADA software?

Q.20 Define friction.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain linear system with example.

Q.22 What are the different types of nonlinearities?

Q.23 What are the advantages and disadvantages of ratio control?

Q.24 Write any five applications of feedback control.

Q.25 What are the differences between inherent and intentional nonlinearities?